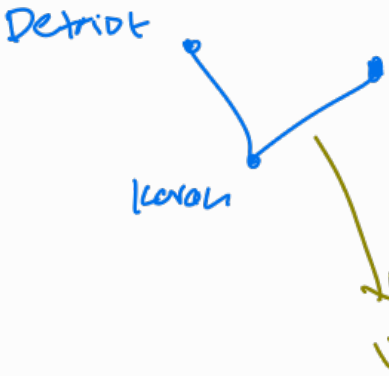
Simple graph (No-directions)

No multiple lines/edges b/w same pair

No loops




vertices/dots/nodes

line connecting is called edge

Pseudograph

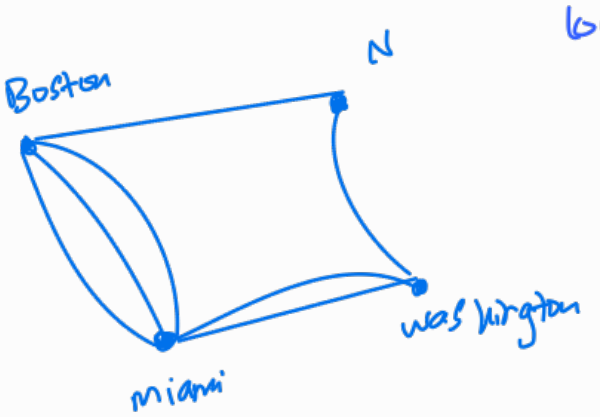
→ can have multiple edges and loops allowed

multigraph

→ can have multiple edges

→ No Loops allowed

But if it is directed then loops allowed




Q1

Boston ⇄ newyork (4, 2)

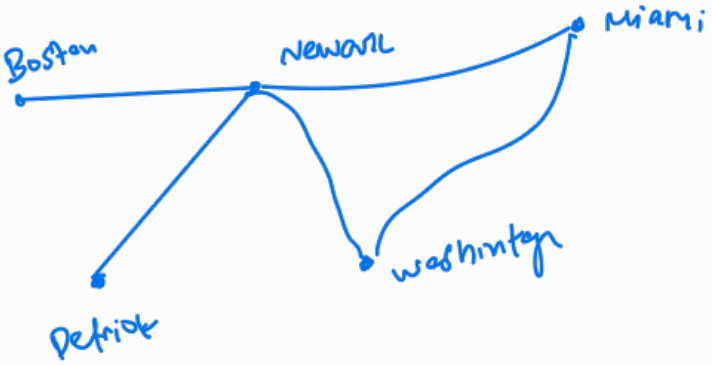
Newark ⇄ Miami (3, 2)

Newark ⇄ Detriot (1, 2)

Newark ⇄ Washington (3, 2)

Washington → Miami (1)

(a)




(b)

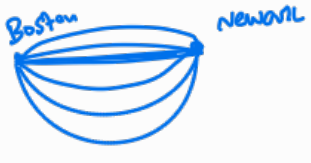



Miami

Washington

Detriot

(c)




Miami

Washington

Detriot

an edge b/w vertices representing cities for each flight

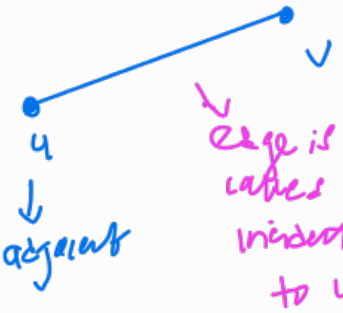
an edge for each flight from a vertex representing a city where the flight begins

fight that operates b/w two in either direction. (undirected)

to where it ends → directed.

Ex 10.2

In an undirected graph → (Simple graph)

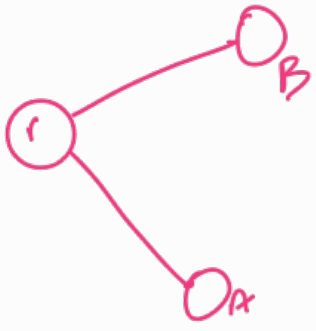



adjacent
u, v are vertices that are called

Degree of vertex in undirected graph:

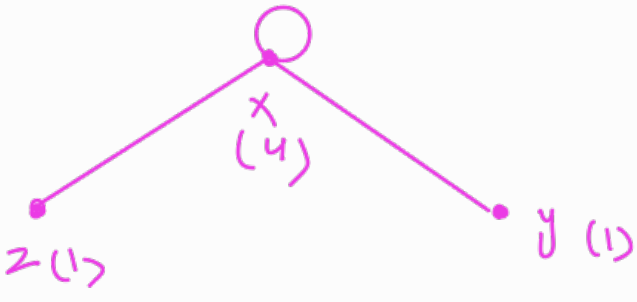
→ is no of edges incident

→ Loop contributes to twice




r neighborhood is A and B.




u(0)
A vertex of degree zero is called isolated

A vertex of degree 1 is called pendent

(Handshaking theorems) → only for undirected graph

Sum of vertex is equal to 2 x no of edges.

(2e)

Q) How many edges are there of a undirected graph.
10 vertices each of degree 6

$10 \times 6 = 2e$

$60 = 2e$

$e = 30$ → edges.

In directed graph only 2

In-degree $deg^-()$ — no. of edges going in

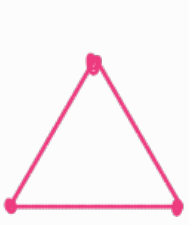
out-degree $deg^+()$ — no. of edge going out

loop → contributes to both in degree and out degree

Sum of $deg^-()$ = Sum of $deg^+()$ → always true only for directed graph

and in a undirected graph the no of vertices with odd degree is always even.

complete graph $(k_n)$:
contains exactly 1 edge btw each pair of distinct vertices





min $k_n = 1$



cycle graph



$C_3$ n → vertices



$C_5$

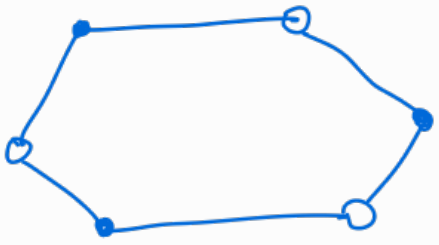
cycle graph min → 3 forms a closed loop

wheels graph → taking the cycle graph and just adding a central node



→ Def:
we split the vertices into two seperate groups. {A, B}, {C, D} and no group cannot connect to itself. A–B ✓ (A–C) (B–D)

Bipartite graph → Always a Simple graph

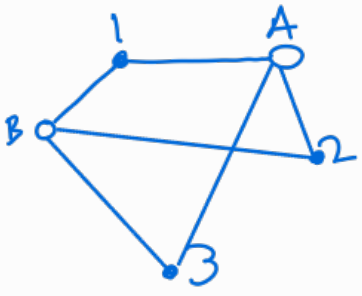


Make bipartite of

V = {1, 2, 3}

U = {A, B}

note. Two bit strings differ exactly by 1 bit if you can change only 1 bit and get the other.




Isomorphic graphs → Two graphs are isomorphic if they are essentially the same but just drawn differently.

graph 1 connecting points A, B, C A → x

graph 2 connecting points x, y, z B → y

C → z

graph 3 connecting points P, Q, R, S (No way to connect)

Two graphs are isomorphic graphs if :

→ every vertex in one graph matches with every vertex another graph.

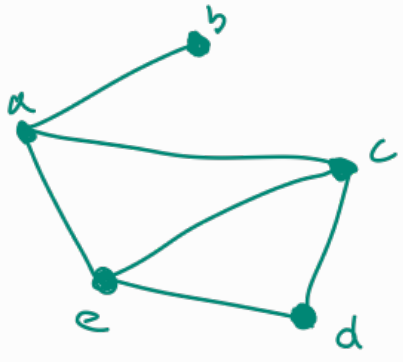
Being isomorphic has equivalence Relation.

- ☆ Reflexive → every graph is iso.- to itself
- ☆ Symmetric → $G_1$ is iso... to $G_2$ then $G_2$ if iso... $G_1$
- ☆ Transitive → $G_1$ is iso... to $G_2$ and $G_2$ is iso... to $G_3$ then $G_1$ is iso ..... to $G_3$

graph Invariants (things that must stay the same) when two graphs are isomorphic

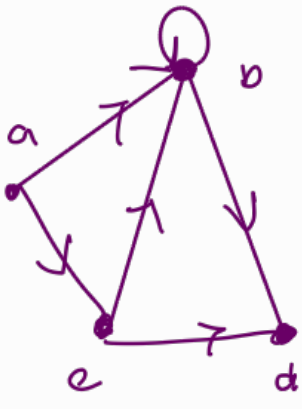
1) No of vertices
2) No of edges
3) degree of vertices

(Adjacency list) ← Representing a graph




which vertex is it connected to ↑

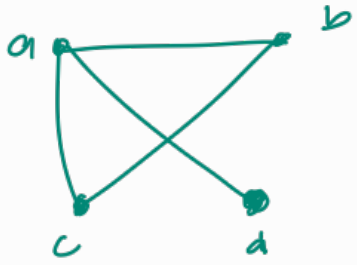
| Vertex | Adjacency list |
|---|---|
| a | b, e, c |
| b | a |
| c | a, d, e |
| d | c, e |
| e | d, c, a |




Adjacency list for directed graph.

| | |
|---|---|
| a | b, e |
| b | b, d |
| d | — |
| e | d, b |

(Adjacency matrix) → tasks which shows which vertices are connected in a graph.

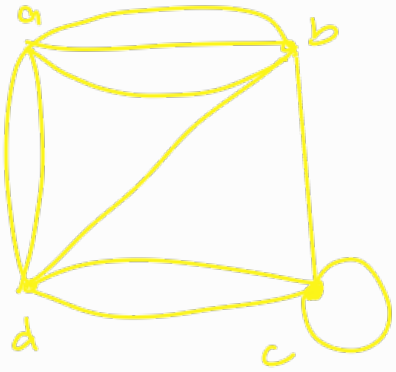



list down over vertices

| | a | b | c | d |
|---|---|---|---|---|
| a | 0 | 1 | 1 | 1 |
| b | 1 | 0 | 1 | 0 |
| c | 1 | 1 | 0 | 0 |
| d | 1 | 0 | 0 | 0 |

1 → edge
0 → N- edge

| | | | |
|---|---|---|---|
| 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 |




multiple edges

| | a | b | c | d |
|---|---|---|---|---|
| a | 0 | 3 | 0 | 2 |
| b | 3 | 0 | 1 | 1 |
| c | 0 | 1 | 1 | 2 |
| d | 2 | 1 | 2 | 0 |

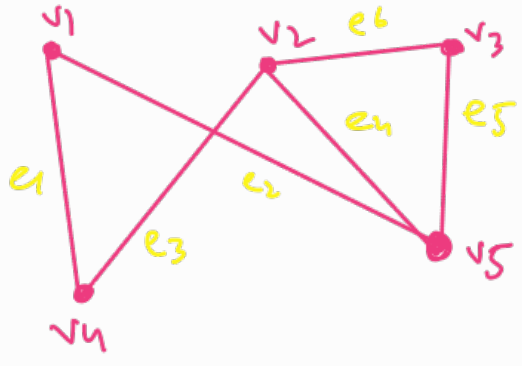
Incidence matrix:-

Predefined

edges

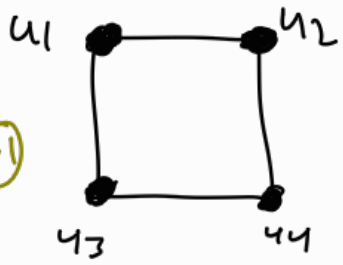
Vertices [ ]

Look column wise and see which edges connect to which vertex place there 1.

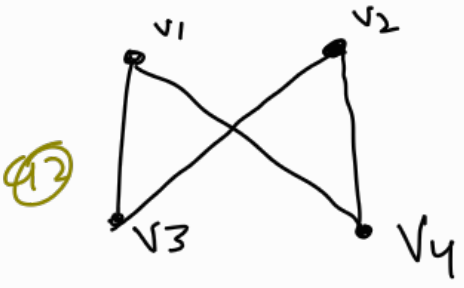



| | $e_1$ | $e_2$ | $e_3$ | $e_4$ | $e_5$ | $e_6$ |
|---|---|---|---|---|---|---|
| $v_1$ | 1 | 1 | 0 | 0 | 0 | 0 |
| $v_2$ | 0 | 0 | 1 | 1 | 0 | 1 |
| $v_3$ | 0 | 0 | 0 | 0 | 1 | 1 |
| $v_4$ | 1 | 0 | 1 | 0 | 0 | 0 |
| $v_5$ | 0 | 1 | 0 | 1 | 1 | 0 |

(Isomorphic graphs)




(G1)




(G2)

$u_1 \to v_1$
$u_2 \to v_2$
$u_3 \to v_3$
$u_4 \to v_4$

edges
$u_1 - u_2$
$u_1 - u_3$
$u_2 - u_4$
$u_3 - u_4$

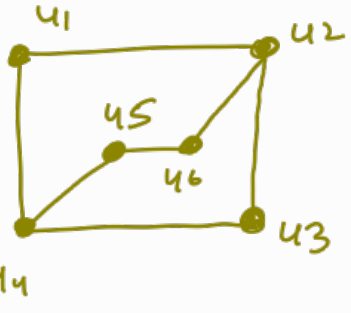
edges
$v_1 - v_3$
$v_1 - v_4$
$v_2 - v_3$
$v_2 - v_4$

To check if graphs are isomorphic first check invariants

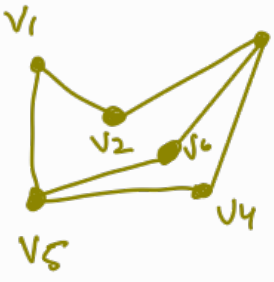
no of vertices
no of edges
degree of vertices

(G1) → 4 (G2) → 4
(G1) → 4 (G2) → 4
(G1) → All vertices degree → 2
(G2) → all vertices degree → 2

If any of them fail graph → not isomorphic but to further 100% confirm this we use mapping.




edges = 7
ver = 6



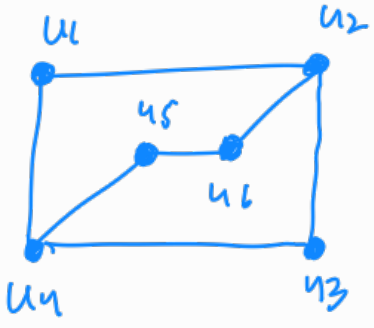

= 7
= 6

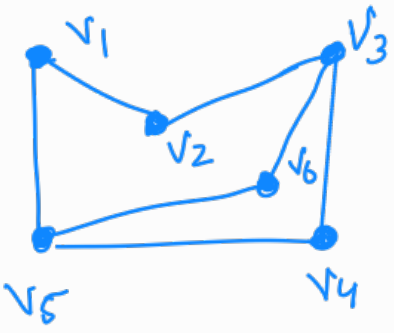
$u_1 \to 2$
$u_2 \to 3$
$u_3 \to 2$
$u_4 \to 3$
$u_5 \to 2$
$u_6 \to 2$

$v_1 \to 2$
$v_2 \to 2$
$v_3 \to 3$
$v_4 \to 2$
$v_5 \to 3$
$v_6 \to 2$

Since it passed all the invariants now we are going to 100% prove if these are isomorphic or not.. using mapping

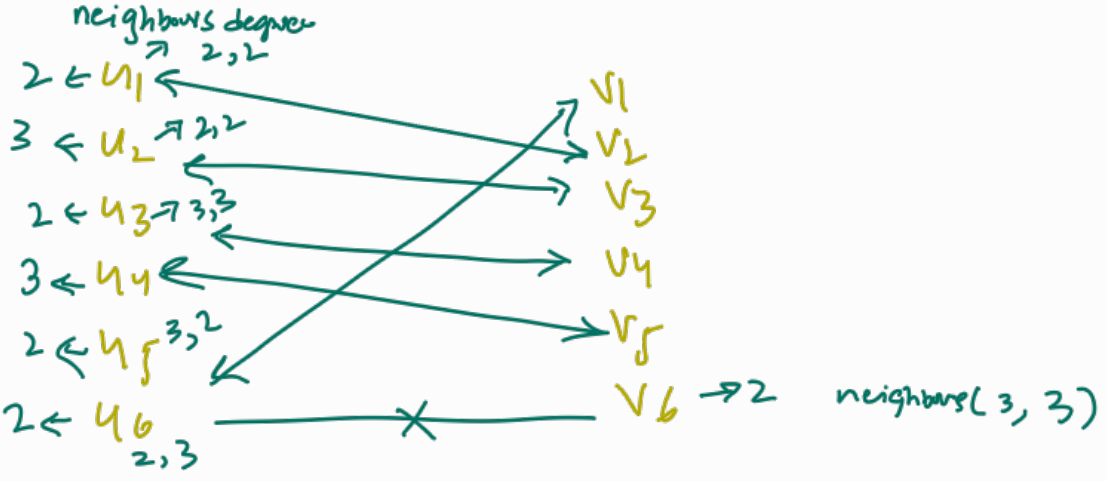
what is isomorphism → When two graphs G and H are same just maybe the vertices are named differently.



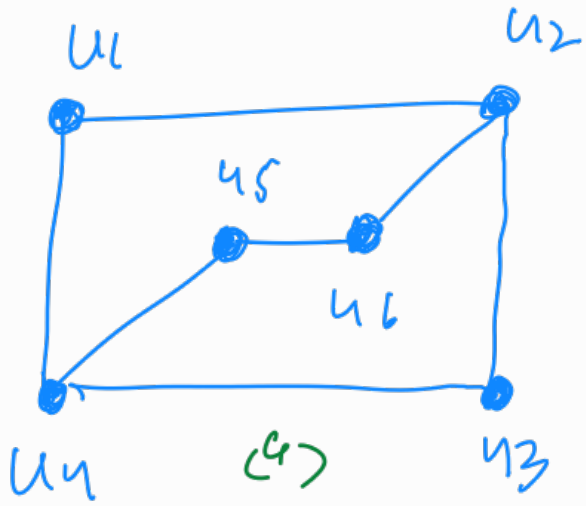



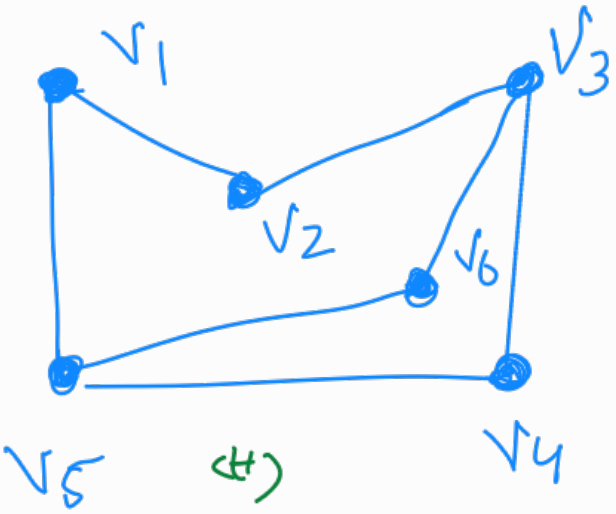

mapping ↓

You match every vertex in graph 1 to exactly 1 vertex in graph 2.




(write degree sequence in decreasing order) ☆



Checking whether the edges are preserved or not.

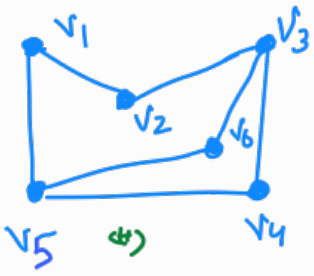
→ Adjacency matrix of G

| | $u_1$ | $u_2$ | $u_3$ | $u_4$ | $u_5$ | $u_6$ |
|---|---|---|---|---|---|---|
| $u_1$ | 0 | 1 | 0 | 1 | 0 | 0 |
| $u_2$ | 1 | 0 | 1 | 0 | 0 | 1 |
| $u_3$ | 0 | 1 | 0 | 1 | 0 | 0 |
| $u_4$ | 1 | 0 | 1 | 0 | 1 | 0 |
| $u_5$ | 0 | 0 | 0 | 1 | 0 | 1 |
| $u_6$ | 0 | 1 | 0 | 0 | 1 | 0 |

| | | | | | |
|---|---|---|---|---|---|
| 0 | 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 | 1 | 0 |

Adjacency matrix of H

| | $v_1$ | $v_2$ | $v_3$ | $v_4$ | $v_5$ | $v_6$ |
|---|---|---|---|---|---|---|
| $v_1$ | 0 | 1 | 0 | 0 | 1 | 0 |
| $v_2$ | 1 | 0 | 1 | 0 | 0 | 0 |
| $v_3$ | 0 | 1 | 0 | 1 | 0 | 1 |
| $v_4$ | 0 | 0 | 1 | 0 | 1 | 0 |
| $v_5$ | 1 | 0 | 0 | 1 | 0 | 1 |
| $v_6$ | 0 | 0 | 1 | 0 | 1 | 0 |

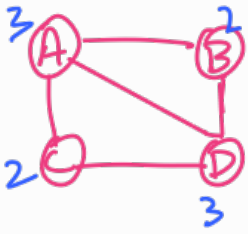



$$\begin{bmatrix} 0&1&0&0&1&0 \\ 1&0&1&0&0&0 \\ 0&1&0&1&0&1 \\ 0&0&1&0&1&0 \\ 1&0&0&1&0&1 \\ 0&0&1&0&1&0 \end{bmatrix}$$

G

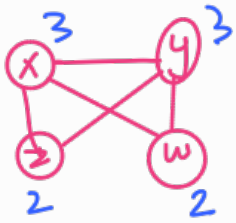
$$\begin{bmatrix} 0&1&0&1&0&0 \\ 1&0&1&0&0&1 \\ 0&1&0&1&0&0 \\ 1&0&1&0&1&0 \\ 0&0&0&1&0&1 \\ 0&1&0&0&1&0 \end{bmatrix}$$

H




3,3,2,2

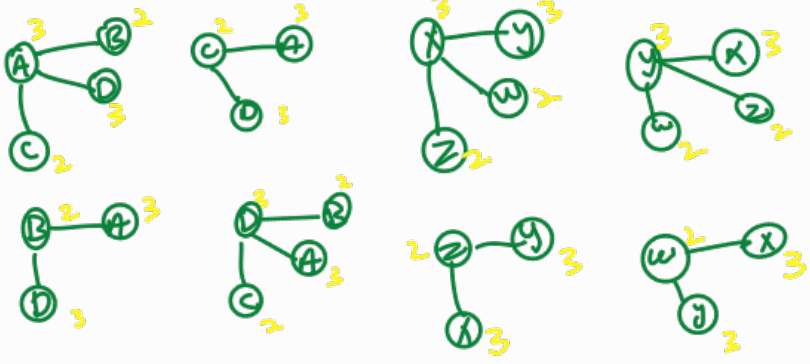
(g1)




3,3,2,2

(g2)

| edges | |
|---|---|
| A-B | x-z |
| A-D | x-y |
| A-C | x-w |
| B-D | z-y |
| C-D | w-y |



A = x
B - z
C - w
D - y

(g1) (g2)

| | A | B | C | D |
|---|---|---|---|---|
| A | | | | |
| B | | | | |
| C | | | | |
| D | | | | |

| | x | z | w | y |
|---|---|---|---|---|
| x | | | | |
| z | | | | |
| w | | | | |
| y | | | | |

must be equal